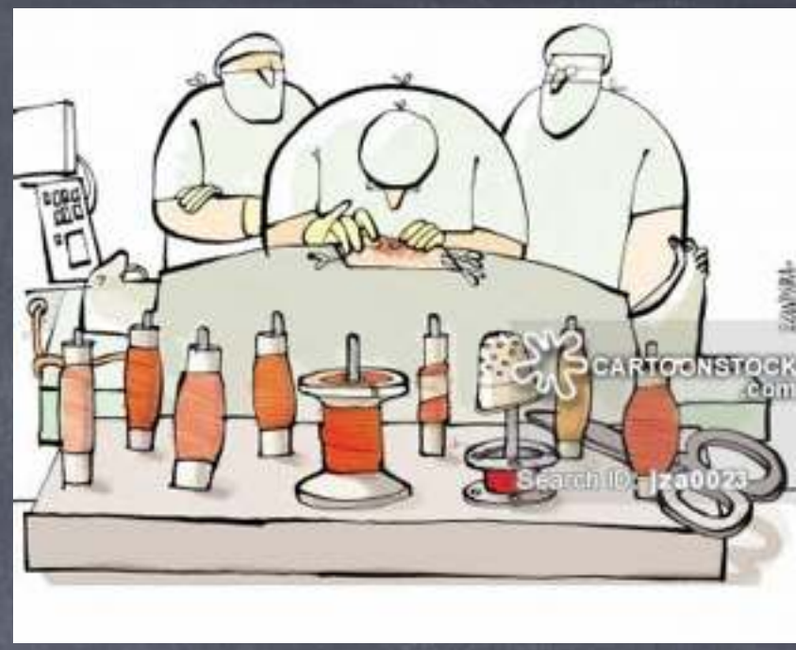
Suture materials, needle, knots and ligatures

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Objective



- Discuss the features, indications, and limitations of the commonly used sutures in large animal surgery.
- Describe the advantages and disadvantages to different types of needles and the indication
- Provide an overview of the indications for and uses of different suture patterns

Suture Material

- Minimal tissue reaction
- Not create a situation favorable for bacterial growth
- Be nonelectrolytic, noncapillary, nonallergenic, and non carcinogenic
- Be comfortable for the surgeon to handle
- Hold knots securely without cutting or fraying
- Disappear as soon as it is no longer needed
- Be economical to use



"I guess you're what they call a snatch purser, huh, Doc?"

Suture material

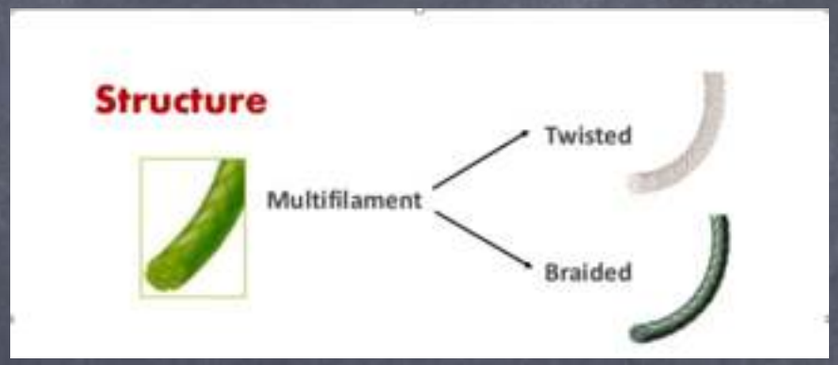
- Absorbable suture
  - Hydrolyzed
  - Disappear when wound healed
- Non-absorbable suture
  - Retain their tensile strength
    - > 60 days



"It's a new technique for training interns: suture by numbers."

Suture material

- Multifilament



  - Harbor bacteria and thus potentiate infection

- Monofilament



  - Synthetic
  - Induce less tissue reaction and exhibit less capillarity

Choice of suture material

- Factors
  - Experience,
  - Knowledge of the healing characteristics of tissues and organs to suture
- Knowledge of the physical and biologic characteristics of various suture materials, and animal factors
  - Age,
  - Weight
  - Overall health status
  - Presence of infection

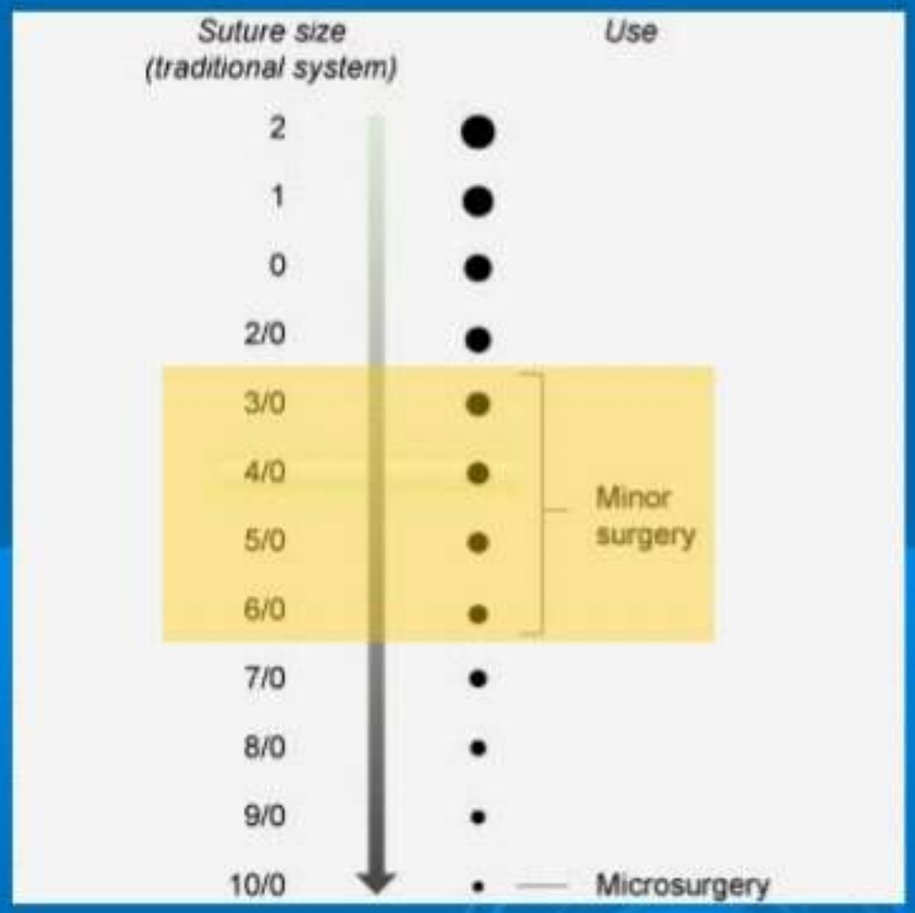
Choice of suture material

- Rate of healing
  - Visceral wound and superficial wound
    - Heal rapidly
- Presence of crystalloid
  - Bladder
    - urinary calculus formation
- Infected wound
  - Chromic gut
    - Local phagocytic activity
  - Braided or multifilament suture
    - Bacterial harboring

Size does matter

- The accepted surgical practice is to use the smallest diameter suture that adequately holds the mending wounded tissue
  - Minimize trauma
  - Tensile strength of the suture need never exceed the tensile strength of the tissue
  - Tissue reaction
  - Compromise blood supply
  - Retard healing

Gauge of the suture materials

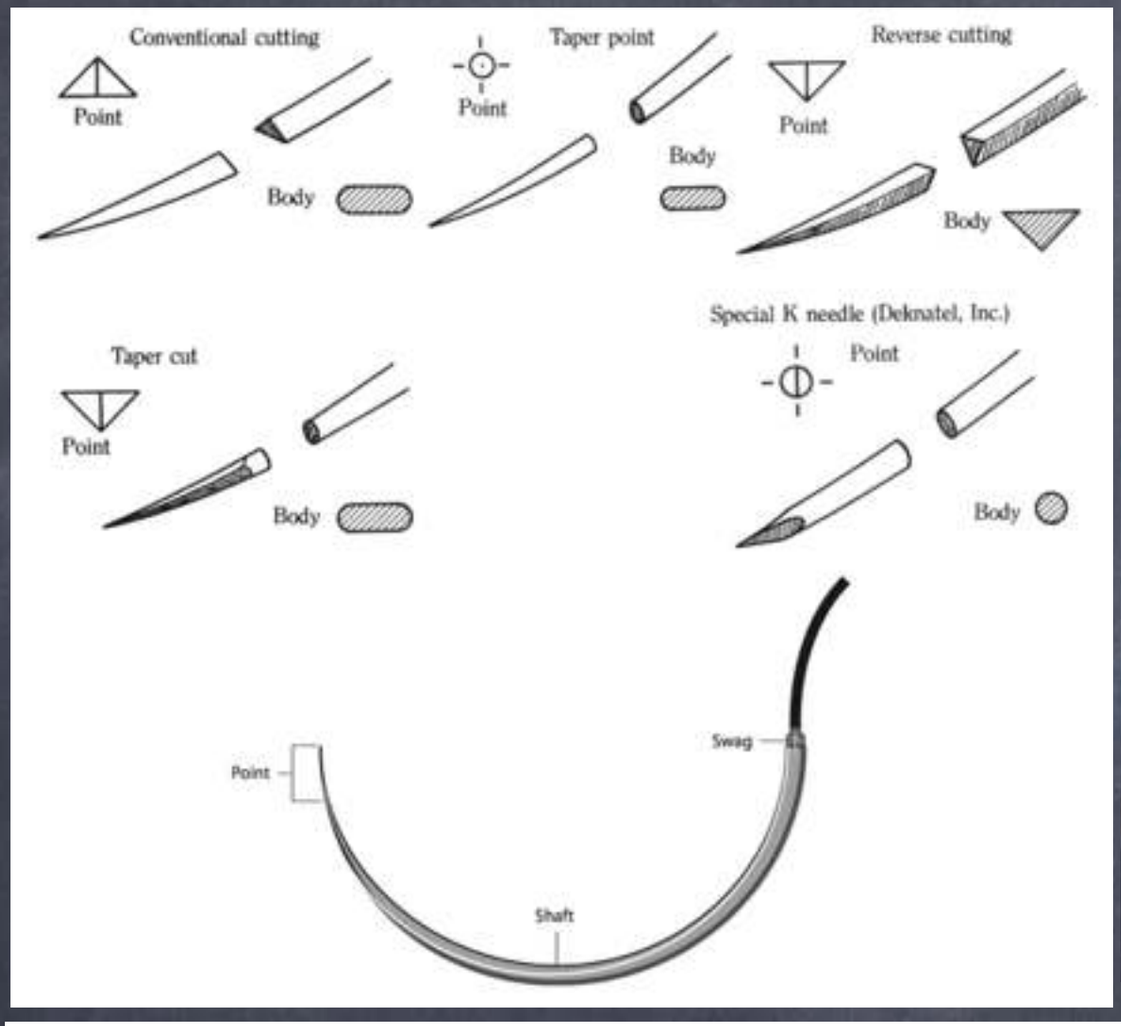


2 Thick
1
0
00
000
0000
00000 Fine

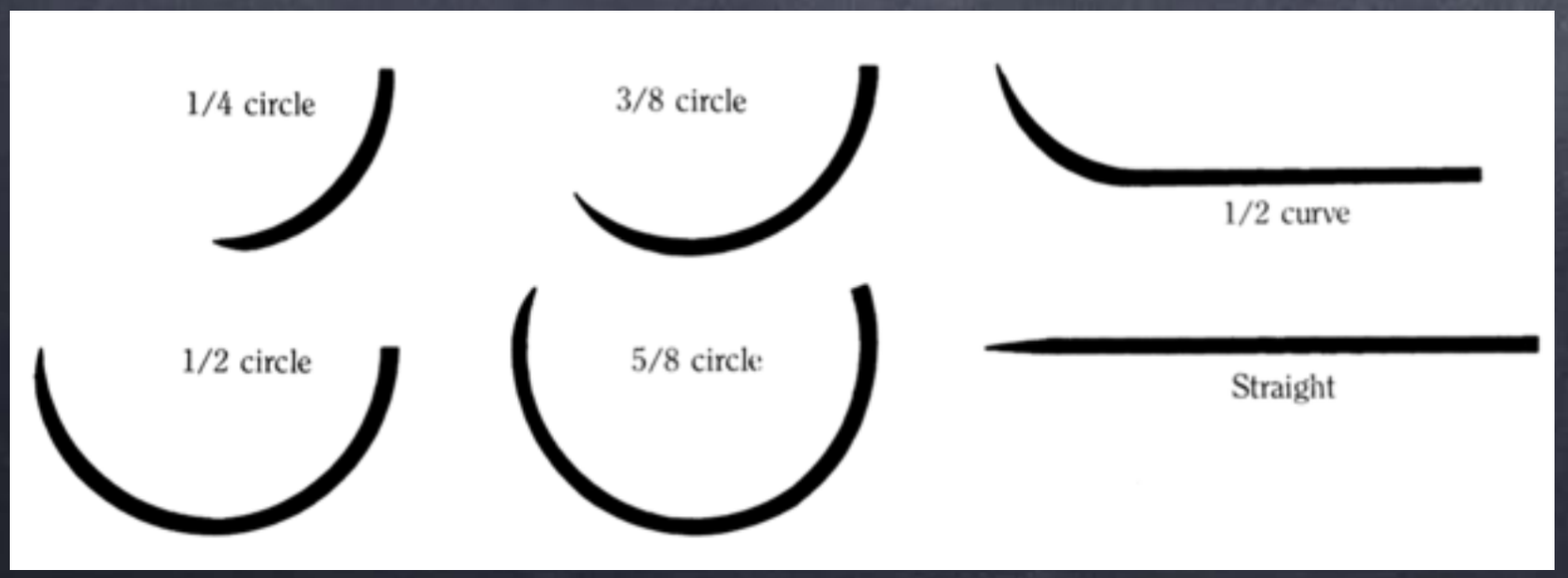
Suture	Material	Qualities	Advantages	Disadvantages
Surgical gut	Natural collagen from the submucosa of sheep intestine	Rapid absorption Digested enzymatically	Economical Good handling characteristics Some elasticity	Knot strength decreases when wet Loses strength within 3–7 days Tissue reaction
Polyglycolic acid (Dexon)	Polymer of glycolic acid	Synthetic Braided Absorbable Hydrolyzed into natural metabolites	Nonantigenic Does not swell when wet Minimal tissue reaction Good knot security Handling characteristics	Some tissue drag Capillary action
Polyglactin 910 (Vicryl)	Polymer of glycolic acid and lactic acid in a ration of 90:10	Synthetic Braided Absorbable Hydrolyzed into natural metabolites	Same as polyglycolic acid	Same as polyglycolic acid
Polyglactin 910 (Vicryl Rapide)	Same as polyglactin 910	Same as polyglactin 910	Same as polyglactin 910 Added benefit of more rapid absorption (50% strength retention at 5 days vs. 50% at 3 weeks for regular Vicryl)	Same as polyglactin 910 Should not be used in areas where extended strength retention is necessary
Polydioxanone (PDS II)	Homopolymer of paradioxanone	Synthetic Monofilament Absorbable Hydrolyzed into natural metabolites	Less tissue drag Does not potentiate infection Persists longer in tissues High breaking strength	Brittle Tendency to break at knots
Polydioxanone antibacterial (PDS Plus Triclosan)	Same as polydioxanone	Same as polydioxanone	Same as polydioxanone Added benefit of providing local antimicrobial effects	Same as polydioxanone
Polyglyconate (Maxon)	Copolymer of trimethylene carbonate and glycolide	Synthetic Monofilament Absorbable Degrades via hydrolysis	Same as polydioxanone	Less breaking strength, greater stiffness, inferior mechanical force compared to PDS

Needles

- Penetrate the tissue with a minimum of resistance and trauma
- Rigid
  - Prevent bending
- Fexible
  - Prevent breaking
- Clean and resistant to corrosion


Conventional cutting
Point
Body
Taper point
Point
Body
Reverse cutting
Point
Body
Taper cut
Point
Body
Special K needle (Deknatel, Inc.)
Point
Body
Point
Swag
Shaft


新华网
WWW.NEWS.CN


1/4 circle
3/8 circle
1/2 curve
1/2 circle
5/8 circle
Straight

Suture Size

Suture Type



Braided (Multi) or
Non-Braided (mono)

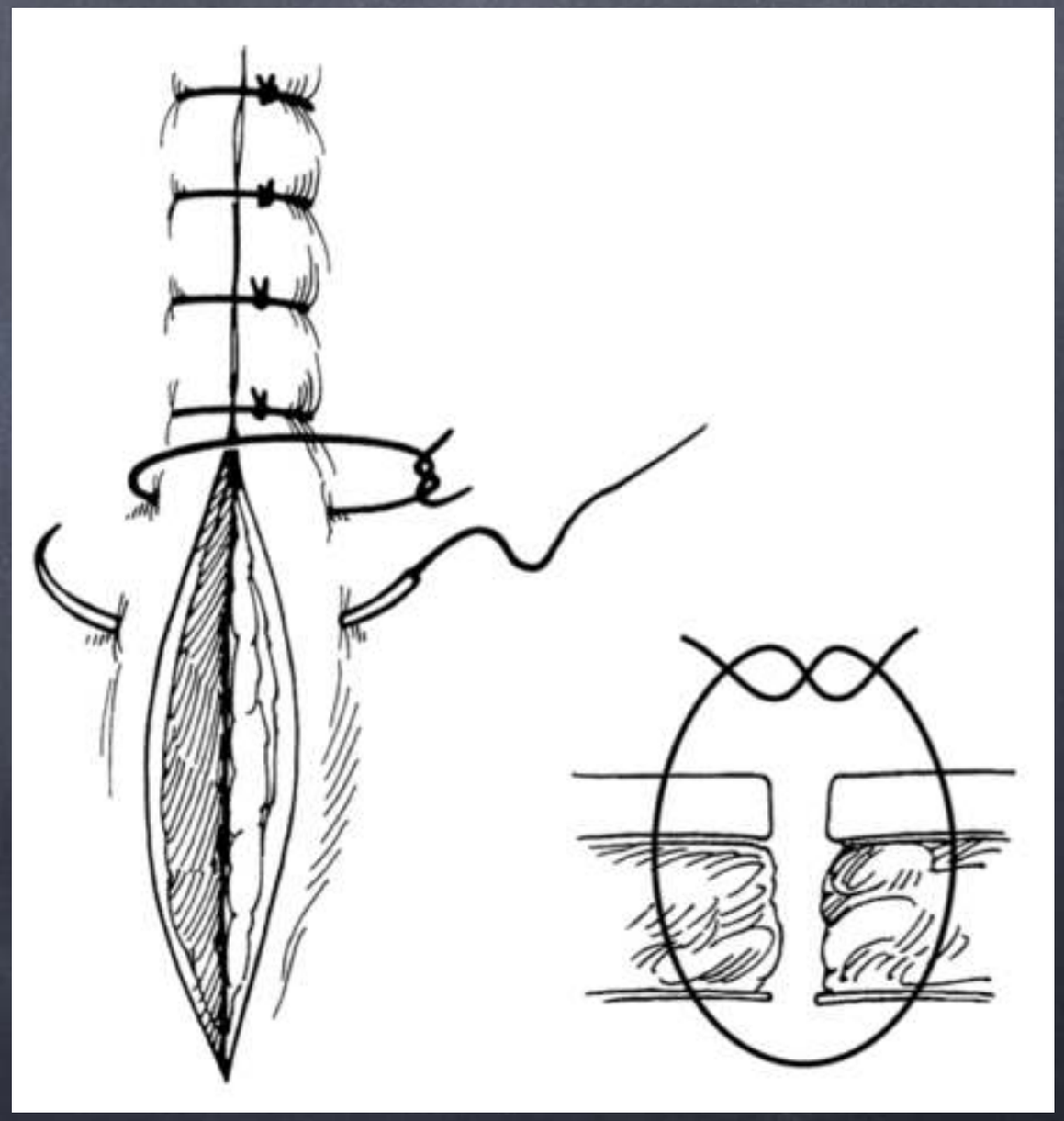
Suture Needle

Natural or Synthetic

Absorbable or Non-Absorbable

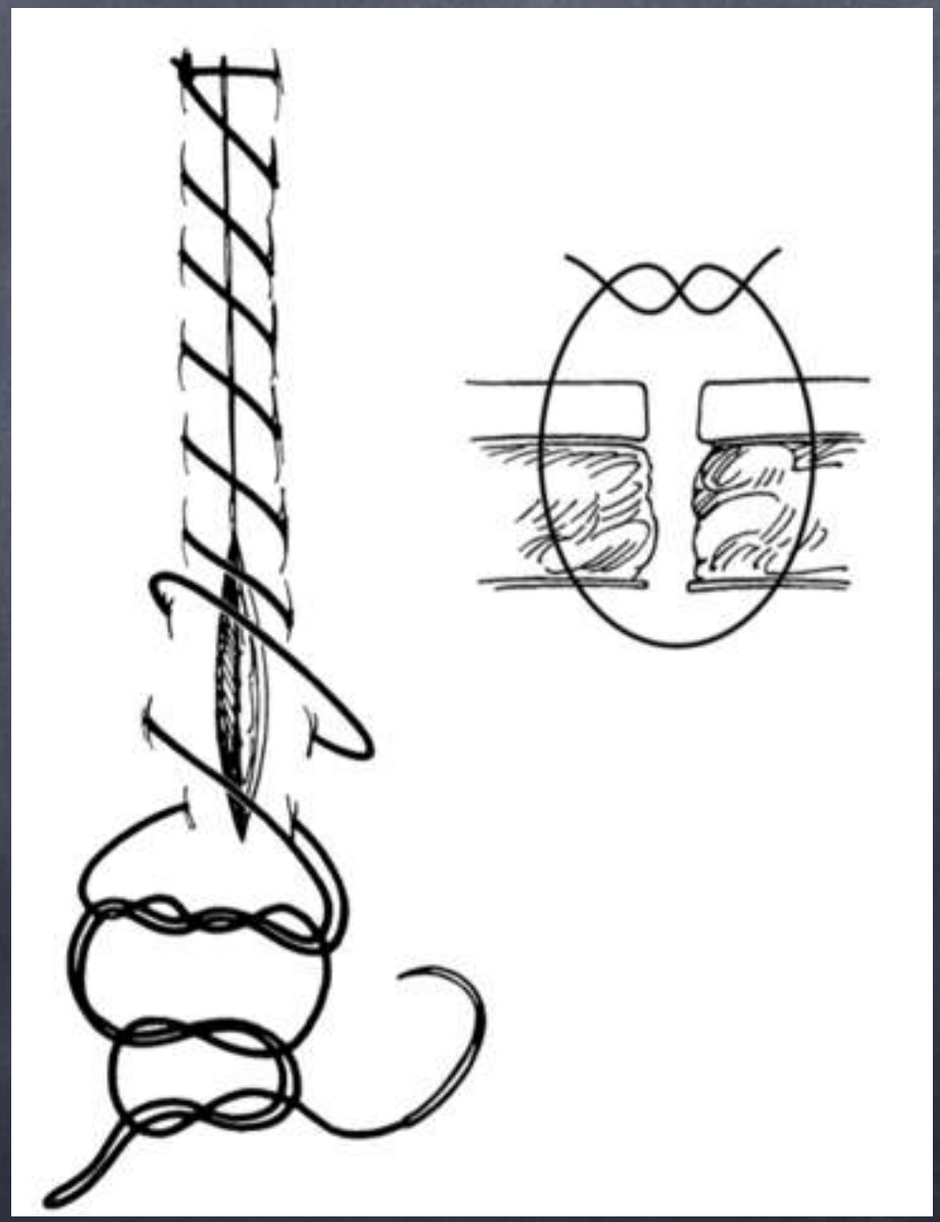
Suture Pattern

- Simple interrupt suture
  - Most widely used
- Appose wound margins
- The distance between sutures is equal to twice the distance between the suture insertion and the wound edge



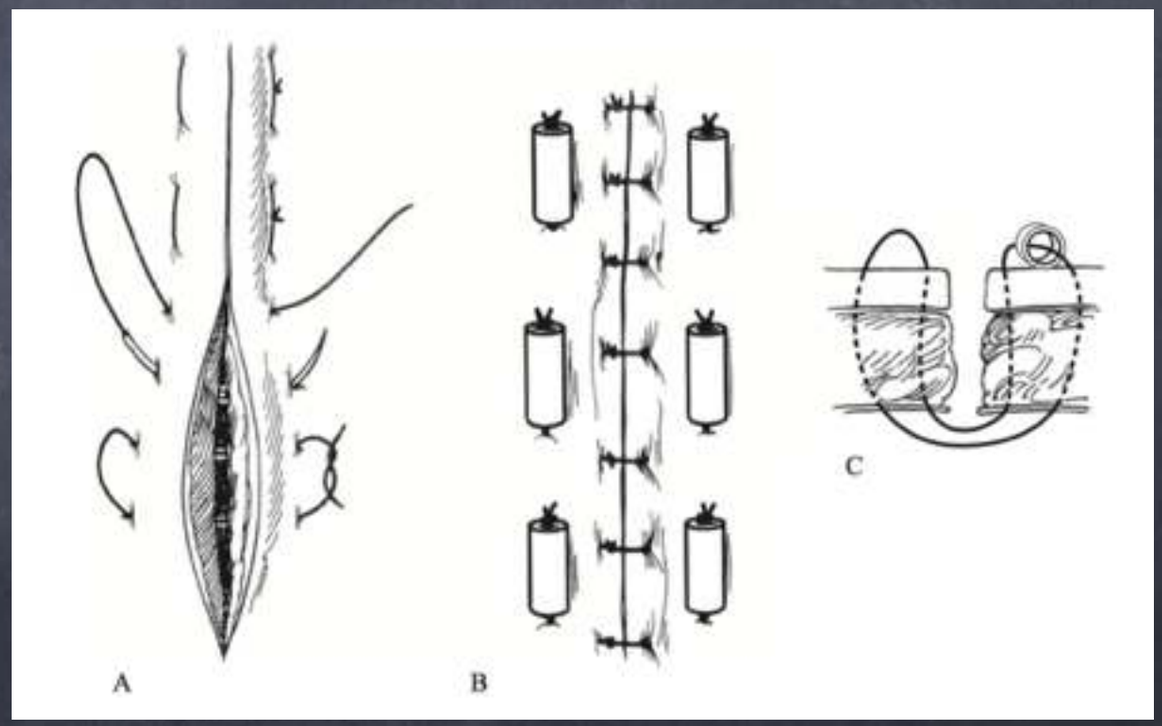
Suture patterns

- Simple continuous suture
  - Elastic suture
  - If any one of the sutures in a continuous suture pattern fails, the strength of the suture line will be lost
- Continuous suture patterns are generally chosen for bowel anastomosis
  - Less foreign material at the anastomosis site
    - Contamination
    - Adhesion
  - More rapidly applied



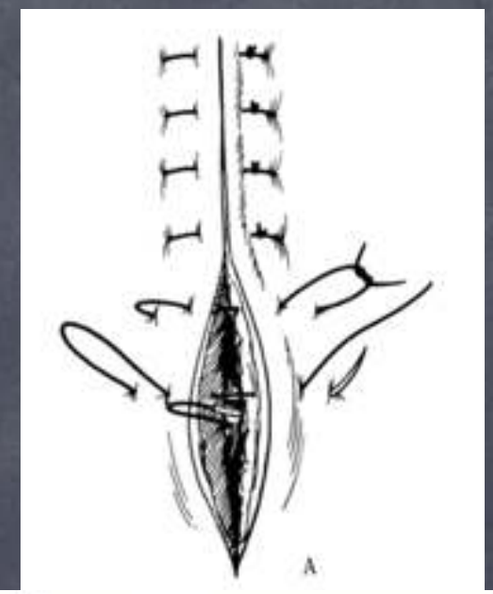
Suture patterns

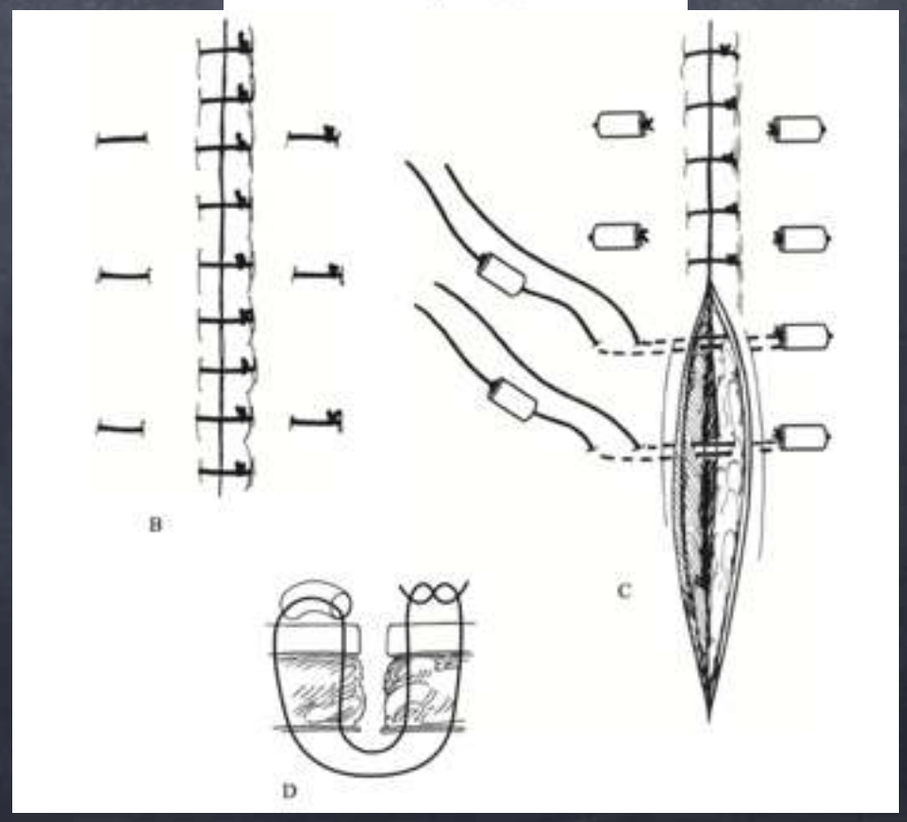
- Interrupt horizontal mattress
  - Everting
  - Best tension reliving suture
    - Reduce blood supply
  - Muscle belly reapposition



Suture patterns

- Vertical mattress suture
  - Better blood circulation
  - Limb laceration





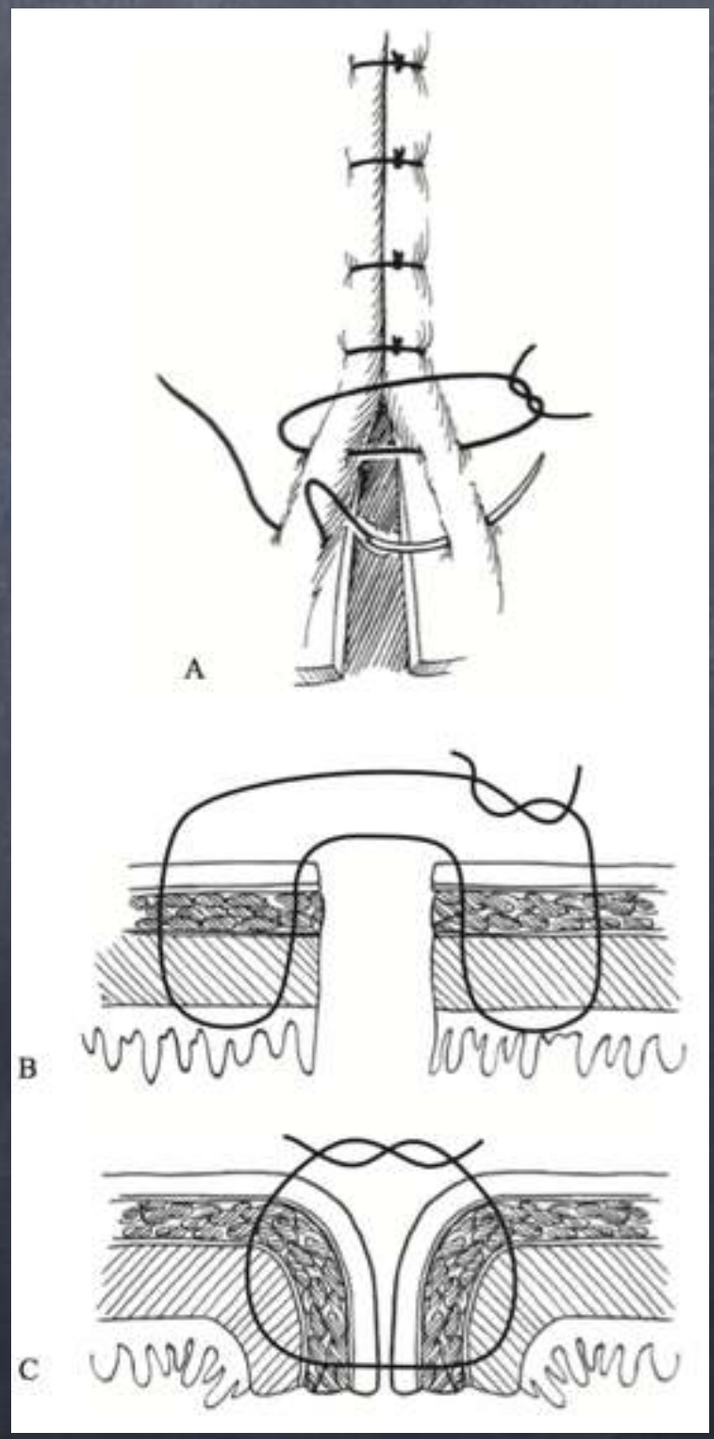
Closure of hollow organ

- Meticulously
  - Inverting or opposing
    - Prevent leakage
    - Fibrin clot
- Strong, pliable GI
- Friable uterus
- Heal quickly



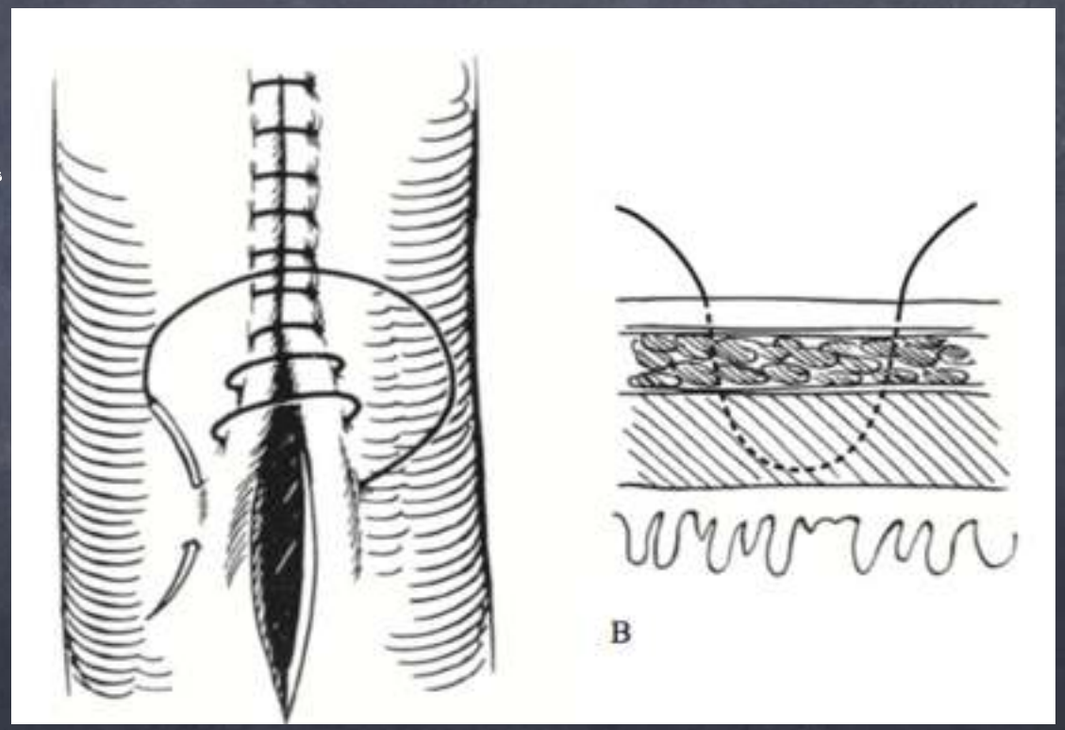
Suture patterns

- Lambert
  - Stomach
    - Rumen
    - Uterus
    - Diameter compromise



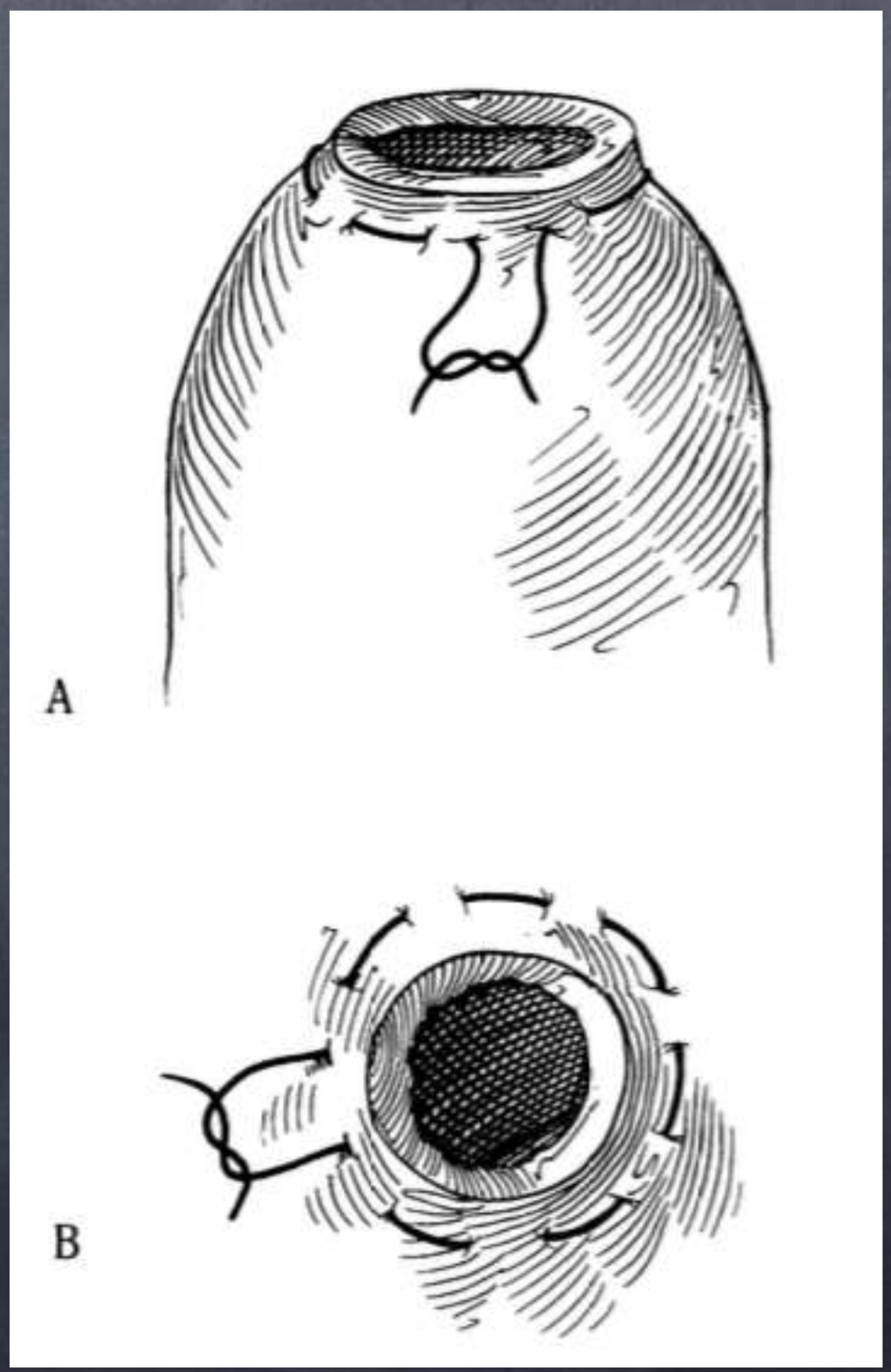
Suture Pattern

- Cushing
  - Minimal inverting
  - Outer layer of double layer closure
  - Blood supply



Suture pattern

- Purse string suture



Suture pattern

- Transfix ligature
- Prevent slipping
  - Friction and tensile
- Leave as little tissue as possible in the stump

